NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

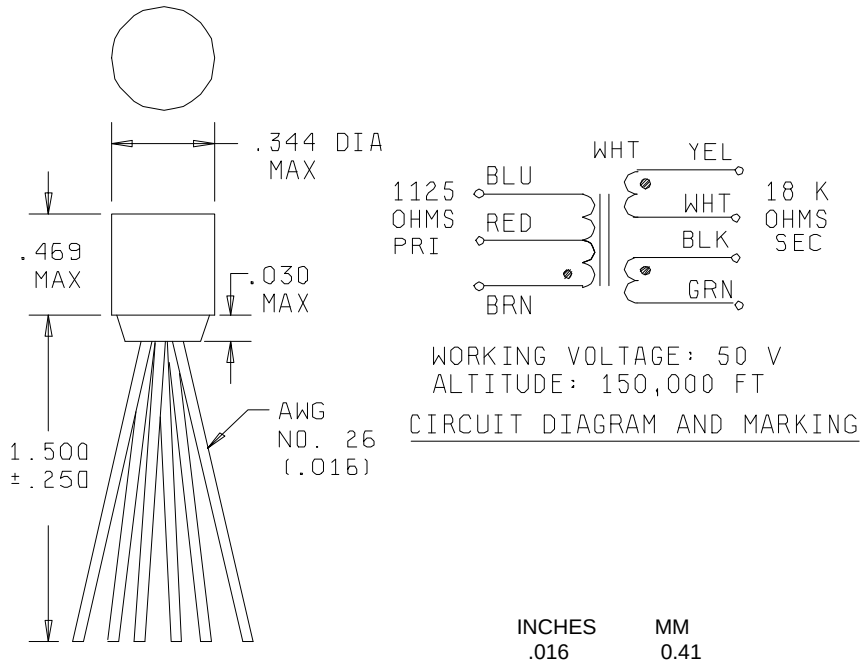
MIL-PRF-27/290A
7 August 1980
SUPERSEDING
MIL-T-27/290
27 March 1980

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY, TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



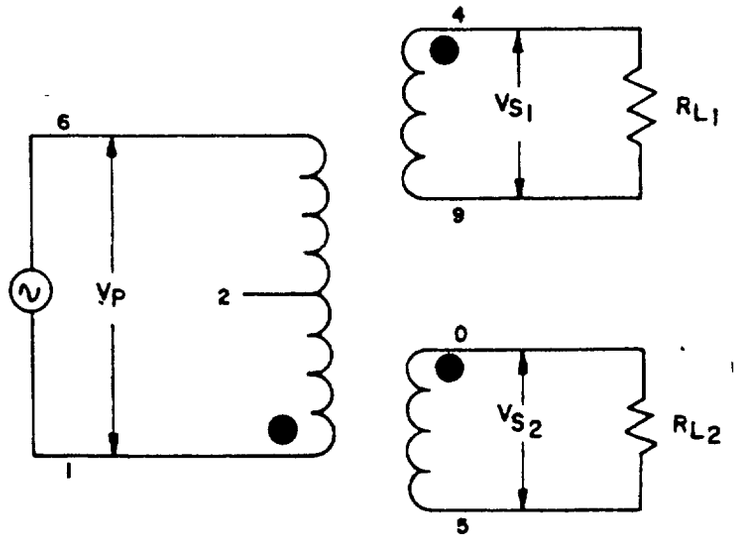
INCHES	MM
.016	0.41
.030	0.76
.25	6.4
.344	8.74
.469	11.91
1.50	38.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

(A) denotes changes

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$V_{S1} = V_{S2} \pm 05\%$ when
$V_p = 15 \sin \{2\pi(400 \pm 20)T\}$ where T = Time in sec
$R_{L1} = R_{L2} = 15\ k\Omega$ thru $1\ M\Omega$

FIGURE 2 Output balance circuit

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REQUIREMENTS. (When colors in parentheses, i.e. (Brn-Blu), are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, i.e (Brn-Blu), the windings are connected in series.)

Electrical ratings.

Ⓐ Primary power level 100 milliwatts maximum at 1 kHz.

Working voltage 50 volts peak

Primary impedance. (Blu-Brn) 1,125 ohms ct ±125 ohms.

Secondary impedance (Yel-Grn) 18,000 ohms split ±2,000 ohms. Split indicates that one-fourth of the listed impedance is available by paralleling the windings.

Primary current 3 5 mA

Frequency range 300 Hz to 20 kHz, ±3 dB at 1 mW level.

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Metal, one open end filled with epoxy

Terminals Copper clad, steel tinned 100 percent, electroplated per MIL-T-10727 and ASTM CCS B452

Diameter 016 inch
Length 1 500 ± 250 inch

Weight 0 2 ounce maximum

Operating temperature range -55° to +130°C

Altitude 150,000 feet

Terminal strength MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding):

At sea level and at reduced barometric pressure 100 volts rms.

Polarity Additive with terminals Blu and Yel, and Wht and Blk connected

Electrical characteristics

Primary impedance 1,125 ohms ct ± 12 percent with 1 volt at 5 kHz and 3 5 mA dc applied to the primary and 18 kΩ across the secondaries

Primary dc resistance 120 ohms ± 25 percent

Secondary dc resistance 940 ohms ± 25 percent.

No load (center tap unbalance). ± 1 percent at 1 volt, 5 kHz across the primary

Harmonic distortion Total harmonic content of the output shall be a maximum of 5 percent at 100 mW and 3.5 mA in the primary.

Secondary output balance When the transformers are operated with identical loads of 15kΩ to 1 MΩ on each of the secondary windings and the primary input voltage is equal to 15 sin [2π (400 ±20) T], where T = time in seconds, the output voltages shall be equal within ±.05% (see figure 2).

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Ⓐ Insertion loss At 100 mW in the primary, the insertion loss shall be a maximum of 3.0 dB at 1 kHz

Frequency response Z_S = 1,000 ohms; Z_L = 16 kΩ, E_L = 1 volt; reference frequency = 1 kHz.

Frequency range is 300 Hz to 20 kHz, ± 3.0 dB

Ⓐ Self-resonant frequency Shall be a minimum of 300 kHz

Vibration (high frequency) MIL-STD-202, method 204, test condition B

Marking location See figure 1

Part number M27/290-01.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, ME
Navy - MC, AS
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0569-26)

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TF5S21ZZ MIL-T-27/290 A TRANSFORMER, AUDIO FREQUENCY

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